

Project Planning Phase

Smart Lender: AI-Powered Loan Approval Prediction System

Project Timeline

Task	Owner	Duration
Planning & Analysis	K Bushra Mehreen	2 Days
Data Preprocessing	K Bushra Mehreen	3 Days
Model Training	K Bushra Mehreen	3 Days
Flask Development	K Bushra Mehreen	3 Days
Testing & Documentation	K Bushra Mehreen	2 Days

Project Milestones

- Project idea finalized.
- Dataset collected and preprocessed.
- Machine Learning models trained and evaluated.
- Gradient Boosting Classifier selected as the final model.
- Flask web application developed.
- System tested successfully.
- Project documentation completed.
- GitHub repository created.

Risk Analysis & Mitigation

- **Imbalanced Dataset:** Balanced using **SMOTE** before model training.
- **Model Overfitting:** Multiple Machine Learning models were evaluated, and the best-performing model was selected.
- **Invalid User Inputs:** Input validation implemented in the Flask web application.
- **Prediction Errors:** Model performance evaluated using a separate testing dataset.

Resources Required

Hardware

- Laptop/Desktop
- Minimum 4 GB RAM
- Internet Connection

Software

- Python
- Jupyter Notebook
- Visual Studio Code
- Flask
- Scikit-learn
- Git & GitHub

Project Deliverables

- Trained Machine Learning Model (.pkl)
- Flask-based Loan Prediction Web Application
- Source Code
- Project Documentation
- GitHub Repository